AAKASH SANGANI

Data Scientist · ML and LLM Systems

Mumbai, India · aakash.sangani2410@gmail.com · linkedin.com/in/aakash-sangani-3149a11a7 · github.com/aakash2410 · aakash2410.github.io

SUMMARY

Data scientist with 3+ years across pharma analytics, fintech experimentation, and AI governance. Builds and evaluates ML and LLM systems, from edge inference and retrieval to Bayesian inference and fairness audits. Author of two open-source AI evaluation tools and a benchmark for agentic failure modes.

TECHNICAL SKILLS

ML and LLM systems: RAG, agentic AI, LLM evaluation and red-teaming, MCP servers, Bayesian inverse reinforcement learning, LSTM, XGBoost, SVM, ensembles, SMOTETomek, SHAP, IBM AIF360

Edge and inference: INT8 quantisation, ONNX, parallel batched inference, sentence-transformers, vector search, ASR / MT / TTS pipelines

Data and experimentation: Python, SQL, pandas, NumPy, PostgreSQL, Apache Spark, A/B testing, scikit-learn, TensorFlow

Infrastructure: AWS, Docker, Ansible, CI/CD gates, SARIF reporting

Governance frameworks: EU AI Act, NIST AI RMF, ISO 42001, GDPR/DPDP, HIPAA, OWASP ASI, MITRE ATLAS

PROFESSIONAL EXPERIENCE

AI Consultant Feb 2026 – present

Kalpa Impact Mumbai

- Quantised the speech and vision models to INT8 via ONNX and restructured inference into parallel batched streams on **Suno Sutra**, the open-source offline AI device from Digital India Bhashini (MeitY) and Current AI: peak memory down 40%, end-to-end latency from ~8s to 2–3s across 22 Indic languages.
- Built and packaged **The Specialist**, a grounded offline retrieval module for that device: multilingual-e5-small embeddings with a deterministic TF-IDF fallback, two refusal gates, and a source citation on every answer. Ships as an additive overlay installed by Ansible, with the corpus refreshed in the device's connectivity window.
- Designed the evaluation framework for the **VYOMA Innovation Challenge** (₹80 lakh, run by Bhashini, Current AI and Kalpa Impact): jury rubric, use-case judging matrix and scoring thresholds, through to the finale at IIIT-Delhi.
- Verified edge-AI performance claims submitted to the Gates Foundation; wrote the reference architecture for AI-enabled coordinated care over ABDM-connected health systems.
- With the Principal Scientific Adviser's Office, wrote a section of *AGI Governance* on AI incident-reporting mechanisms India can adopt, drawing on the EU AI Act and California's SB-53.

Business Technology Solutions Associate → Senior Associate Sep 2023 – Jan 2026

ZS Associates Pune

- Launched an agentic AI analytics product for natural-language querying of sales-performance metrics.
- Owned incentive-compensation product logic for a Fortune 100 pharma client, delivered at 100% accuracy.
- Led a team of 4. Recognised as a top 1% performer firm-wide; promoted to Senior Associate within 8–10 months.

Product Analyst Intern Jul 2022 – Feb 2023

Bright Money Bangalore

- Launched a referral programme that saved $120K in customer-acquisition cost over 5 months.
- Built A/B testing infrastructure and SQL dashboards.

SELECTED PROJECTS

Inverse-Objective Regulator `github.com/aakash2410/Inverse-Objective-Regulator`

- Black-box auditor that infers a deployed agent's revealed objective from its tool-call traces via Bayesian inverse reinforcement learning, scores divergence from the declared purpose, and emits evidence mapped to the EU AI Act, OWASP, MITRE ATLAS and NIST AI RMF as JSON and SARIF.
- Open 8-agent benchmark of agentic failure modes (reward hacking, power-seeking, sycophancy, privacy leakage, safety-skipping). A deterministic open cross-encoder critic recovered planted objective divergence on 6 of 8 agents (mean r = 0.55), beating a frontier LLM-as-judge (0.08 → 0.43) with near-zero variance and no API cost. Sole author of the accompanying arXiv system paper.

rag-compliance-auditor `github.com/aakash2410/MCP_RAG_Compliance`

- Sole-authored open-source package on PyPI. MCP server that certifies RAG and AI pipelines against the EU AI Act, GDPR/DPDP, HIPAA, SEBI/RBI and NIST AI RMF across 7 audit dimensions.
- Issues RS256-signed JWT certificates, exposes 8 MCP tools, and runs as a CI/CD gate with SARIF reporting.

Credit Equity Audit of India `github.com/aakash2410/CreditAudit-India`

- Two-stage, Heckman-style deep learning model on the NSSO 77th Round separating access to the credit market from allocation of formal versus informal credit. Female-headed households had ∼59% of the probability of market entry of male-headed households (disparate impact 0.59).
- IBM AIF360 adversarial debiasing and SHAP to test for proxy discrimination, reaching near-ideal disparate impact at ∼73.2% accuracy. Documented the design's limits: self-reported survey data, collider bias risk, and SHAP as a model diagnostic rather than causal evidence.

Earlier work 2022 – 2023

- **Baby Cry Reason Detection.** Cause of an infant's cry from audio: MFCC, spectral contrast and prosodic features over 457 samples, SMOTETomek for class imbalance, soft-voting SVM + KNN ensemble, deployed in a Flutter app with a retraining feedback loop. Team of 3.
- **Personality Prediction from CV and Social Media.** A 7-layer LSTM reached 0.82–0.85 accuracy on the Big Five against 0.45–0.61 for SVM, logistic regression, gradient boosting and XGBoost (myPersonality: 250 users, 9,917 status updates). Team of 4.

PUBLICATION

Sangani, A., Doshi, P., & Sawant, V. "Unusual Social Media Behavior Detection Using Distributed Data Stream Mining." In *Practical Data Mining Techniques and Applications*. Auerbach Publications (CRC Press), 2022–23. Clickstream-based bot and hijack detection using per-cluster Very Fast Decision Trees on Apache Spark, with concept-drift handling.

EDUCATION

B.Tech, Information Technology 2019 – 2023

D. J. Sanghvi College of Engineering, University of Mumbai CGPA 8.90 / 10

- Merit scholarship covering 100% of tuition. Coursework: Probability & Statistics (O, 93%), Advanced Data Structures (O, 95%), Artificial Intelligence (A+, 84%), Statistical Analysis (A+, 81%), Design & Analysis of Algorithms (A+, 83%), Discrete Structures (A+, 84%).